Shan Yang

Senior Applied Scientist — Foundation Models, Multimodal AI, Generative AI

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Professional Summary

- Technical leader with 8+ years driving foundation model research and production ML systems at scale, spanning multimodal generative AI, video/image generation, and LLM post-training across Amazon and Google
- Proven track record leading teams and cross-functional initiatives, managing Applied Scientists, and establishing technical roadmaps that advance state-of-the-art research while shipping products serving millions of users
- Deep expertise in image/video/3D generation, multimodal foundation models (vision-language), with pioneering work in music-conditioned motion synthesis and attention-based multimodal fusion (800+ citations)

Education

University of North Carolina at Chapel Hill — Chapel Hill, NC
Ph.D., Computer Science — 2012 – 2017

Shanghai Jiao Tong University — Shanghai, China
B.E., Software Engineering — 2008 – 2012

Technical Skills

ML Frameworks: PyTorch, TensorFlow, Scikit-learn — Distributed training (DeepSpeed, FSDP)

Foundation Models: Diffusion Models, Multimodal LLMs, Transformers, Attention Mechanisms, CLIP, BLIP

Research Focus: Novel Model Architectures, Image/Video/3D Generation & Editing, Multimodal Understanding, Post-Training (RLHF), Test-Time Training, Reinforcement Learning

Architecture Innovation: Attention Bottlenecks, Multimodal Fusion, Music-Conditioned Motion Synthesis, Autoregressive Diffusion, Linear Recurrent Models

Engineering & Research Experience

Amazon Prime Video — Sunnyvale, CA
Senior Applied Scientist Tech Lead — June 2025 – Present

- Leading **GenAI Live Action Studio** (image/video generation products) and team of **5 Applied Scientists**, researching novel autoregressive diffusion architectures and establishing scalable training infrastructure supporting 10B+ parameter models
- Driving technical innovation in controllable generation through novel autoregressive diffusion models and post-training techniques (**Test-Time Training**, Teacher-Diffusion-Forcing, RL), advancing state-of-the-art in generative models
- Leading cross-functional initiatives with live action directors and creative teams to improve creative workflows, translating artistic requirements into novel architecture designs for production-ready controllable generation systems

Amazon Search — Palo Alto, CA
Senior Applied Scientist Tech Lead — September 2021 – June 2025

- Led end-to-end development of **Amazon Video Search** from zero to launch serving millions of customers, managing **3 Applied Scientists and 2 MLEs**, influencing SDE team of 15: established multi-year technical roadmap, navigated ambiguity space through data-driven architecture decisions, designed novel multimodal ranker, and established production ML systems
- Pioneered multimodal fusion architecture integrating video, text, image, and structured data through iterative experimentation and ablation studies, making critical design choices that enabled multi-modal search at scale
- Drove research agenda in multimodal LLM robustness, developing **Schema-of-Thought** prompting framework achieving **+28% hallucination reduction** and improved out-of-distribution generalization, demonstrating technical leadership in LLM robustness and safety
- Led cross-functional product initiatives across engineering and product teams, filed multiple patents, and published at premier venues (ECCV, AAAI), advancing both research and production impact

Google Research Mountain View, CA
Senior Research Software Engineer June 2017 – September 2021

- Led foundational research in multimodal generative models, influencing **3 Research Engineers**, driving research agenda in music-conditioned 3D motion synthesis and video-audio-motion modeling, publishing at premier ML conferences (ICCV, NeurIPS, ECCV)
- Drove technical ownership of **Google's production 2D/3D reconstruction models** deployed on **GCP** at scale, demonstrating ability to lead end-to-end ML product development from research to production
- Provided technical leadership through patent contributions, cross-functional collaboration, and mentorship of research interns on advanced ML projects

Research in Progress

Schema-of-Thought: Prompting framework for multimodal LLM robustness and out-of-distribution generalization (+28% Claude 3.5, +13.3% GPT-4o)

Linear Recurrent Models for Test-Time Reasoning: Novel architecture leveraging LQR optimal control theory to improve LLM reasoning (+35.8% on Math-500, +10% on AIME25)

Teacher Diffusion Forcing: Novel training paradigm merging teacher forcing with diffusion forcing for autoregressive diffusion, modeling error as conditional distribution

ClipDreamer: Subject-driven video generation using test-time training to inject long video clip information for improved subject fidelity

Selected Publications

ViLA: Efficient Video-Language Alignment for Video Question Answering
X Wang, J Liang, CK Wang, K Deng, Y Lou, MC Lin, ***S Yang***
ECCV 2024
Efficient vision-language multimodal foundation models.

ICAR: Image-based Complementary Auto Reasoning
X Wang, A Liang, J Liang, M Lin, Y Lou, ***S Yang***
AAAI 2024
Multimodal reasoning for visual understanding tasks.

Attention Bottlenecks for Multimodal Fusion
A Nagrani, ***S Yang****, A Arnab, A Jansen, C Schmid, C Sun*
NeurIPS 2021 — **800+ citations**
Multimodal fusion architecture for foundation models.

AI Choreographer: Music Conditioned 3D Dance Generation with AIST++
R Li,* ***S Yang****, DA Ross, A Kanazawa*
ICCV 2021 — **700+ citations**
Multimodal generative modeling: music, motion, vision.

Physics-Inspired Garment Recovery from a Single-View Image
S Yang*, Z Pan, T Amert, K Wang, L Yu, T Berg, MC Lin*
ACM ToG 2018 — **100+ citations**
3D generation from images.

Modeling Context in Referring Expressions
L Yu, P Poirson, ***S Yang****, A Berg, T Berg*
ECCV 2016 — **1600+ citations**
Vision-language understanding.

Patents & Intellectual Property

VLAP: Efficient Video-Language Alignment via Frame Prompting and Temporal Distilling (2023)

Monitoring Animal Pose Dynamics from Monocular Images (2022)

Image-based Complementary Item Recommendation Algorithm (2022)

Auto Design: ML-based Furniture Layout Recommendation Algorithm (2022)